

Slide 1: Relational Model Fundamentals

23IT201 • Database Management Systems • Module 4: Relational Model & Relational Algebra

A relation is a table with rows (tuples) and columns (attributes). Candidate keys uniquely identify tuples; foreign keys reference primary keys.

Slide 2: Basic Relational Algebra Operations

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Selection (σ) filters rows; Projection (π) selects columns; Cartesian Product ($\times$) combines all tuple pairs.

Slide 3: Natural Join & Set Operations

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Natural Join ($\bowtie$) pairs tuples based on matching attribute names. Union ($\cup$), Intersection ($\cap$), and Set Difference ($-$).